Probabilistic and Temporal Machine Learning Approaches for 5G Handover Prediction and Analysis

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***Abstract*—Fifth-generation (5G) cellular networks must maintain service continuity as mobile user equipment moves across radio coverage regions. Handover management consequently requires timely identification of mobility-related warning conditions, selection of a suitable target serving cell, and anticipation of future handover activity. This paper presents a probabilistic and temporal machine learning framework for analysing these tasks using the Krakow handover dataset from the FUMD-AI repository. Four simulation runs are combined, resulting in 1,408,271 observations and 28 retained data columns. A feature representation comprising 15 numerical variables and 42 one-hot encoded serving-cell indicators is constructed.**

The framework addresses three learning tasks: (i) binary prediction of the dataset-provided migration warning, (ii) destination serving-cell classification for warning-positive observations, and (iii) regression of future serving-cell transition counts over 30- and 60-second horizons. In addition, serving-cell transitions are used to derive cumulative handover counts, historical transition counts, and a past handover rate. Gaussian Naive Bayes, linear and quadratic discriminant analysis (LDA and QDA), logistic regression, a Gaussian mixture model (GMM), and a Laplace-approximated Bayesian logistic model are evaluated for classification. Poisson regression and histogram gradient boosting with Poisson loss are evaluated for future event-count prediction. Vehicle-group-aware data partitions are used to reduce leakage between observations from the same simulated vehicle.

On the recorded test set, Bayesian logistic regression achieves the highest binary PR-AUC (0.2194), while QDA achieves the highest F2-score (0.4348). Logistic regression gives the strongest destination classification result, with a macro-F1 score of 0.9024. Histogram gradient boosting outperforms Poisson regression for both future-count horizons, obtaining test MAE values of 0.4261 and 0.8012 for 30 and 60 seconds, respectively. The results demonstrate the usefulness of probabilistic classification and nonlinear count regression while highlighting class imbalance and limitations that must be addressed before deployment.

***Index Terms*—5G mobility management, handover prediction, serving-cell selection, temporal event counting, Gaussian mixture model, Bayesian logistic regression, Poisson regression, gradient boosting.**

I. Introduction

A cellular network partitions a geographical service area into cells, each served by a base station or radio access node. In 5G New Radio (NR), the radio access network coordinates radio-resource allocation, measurements, mobility procedures, and connectivity between user equipment (UE) and the network. As a UE moves, the radio link to its current serving cell may weaken while a neighbouring cell becomes more suitable. A handover transfers the UE connection from the source serving cell to a target cell while attempting to preserve ongoing communication. The 5G NR architecture and associated radio procedures are specified by 3GPP [1], [2].

Handover is essential for coverage continuity, but its timing and frequency affect network performance. A late handover can lead to radio-link degradation, interruption, or failure. Conversely, unnecessary or frequent handovers can increase signalling overhead, cause ping-pong behaviour, and reduce resource efficiency. A useful mobility-management system should therefore provide more than a binary warning. It may need to estimate whether a warning condition is present, identify a likely destination cell, quantify recent handover activity, and forecast the number of transitions expected within a future time interval.

Radio quality, mobility, network performance, and serving-cell history interact in potentially nonlinear ways. Pattern recognition and machine learning methods can learn statistical relationships between these measurements and handover-related outcomes. Generative probabilistic methods such as Naive Bayes, discriminant analysis, and GMMs model class-conditional distributions. Discriminative methods such as logistic regression estimate conditional class probabilities directly. Bayesian logistic regression additionally represents uncertainty in model parameters. For targets representing numbers of events, count regression is more appropriate than categorical classification.

This work develops a multi-task pipeline for 5G handover analysis using four Krakow simulation runs. The main contributions are:

- binary prediction of the dataset-provided migration warning;

- destination serving-cell prediction conditioned on a positive migration warning;
- derivation of cumulative and recent serving-cell transition counts;
- estimation of historical handover rate in events per minute; and
- prediction of future transition counts over 30- and 60-second horizons.

An important distinction is maintained throughout the work: the migration warning is a supplied dataset label, whereas a transition event is independently derived by comparing consecutive serving-cell observations.

II. Problem Formulation

Let observation i belong to vehicle group g_i, be recorded at time t_i, and have feature vector $\boldsymbol{x}_i \in \mathbb{R}^d$. Let m_i denote the migration label, c_i the current serving cell, and d_i the destination label.

A. Binary Migration-Warning Prediction

The binary target is defined from the migration field as

$$y_i^{\text{warn}} = \mathbb{1}(m_i > 0) = \begin{cases} 1, & m_i > 0, \\ 0, & m_i = 0. \end{cases} \tag{1}$$

A classifier estimates $P(y_i^{\text{warn}} = 1 \mid \boldsymbol{x}_i)$. A probability threshold τ gives the final decision:

$$\hat{y}_i = \mathbb{1}(\hat{p}_i \geq \tau). \tag{2}$$

The threshold is selected using validation F2-score and then applied unchanged to the test set.

B. Destination Serving-Cell Classification

For warning-positive observations, the target is the categorical destination $d_i \in \mathcal{C}$, where the observed destination classes are $\mathcal{C} = \{1, \ldots, 9\}$. The predicted destination is

$$\hat{d}_i = \arg\max_{c \in \mathcal{C}} P(d_i = c \mid \boldsymbol{x}_i, y_i^{\text{warn}} = 1). \tag{3}$$

Destination models are trained and evaluated only on rows for which $m_i > 0$.

C. Serving-Cell Transition Events

The temporal analysis defines a transition event by comparing consecutive serving-cell observations within each vehicle:

$$H_{g,j} = \mathbb{1}(C_{g,j} \neq C_{g,j-1}), \qquad H_{g,1} = 0. \tag{4}$$

The cumulative transition count at time t is

$$N_{\text{cum},g}(t) = \sum_{j : t_{g,j} \leq t} H_{g,j}. \tag{5}$$

This is a derived descriptive quantity; a separate model is not trained to predict cumulative count.

TABLE I
Summary of the project objectives

Task	Input information	Output
Warning prediction	57-dimensional feature vector	Migration-warning probability
Destination prediction	Warning-positive observations	Destination cell in $\{1, \ldots, 9\}$
Cumulative count	Serving-cell sequence	Transitions up to current time
Past count/rate	Serving-cell sequence and time window	Recent count and events/min
Future count	Current features and history	Count in next 30/60 s

D. Historical Counts and Handover Rate

For a window length $T \in \{30, 60\}$ seconds, the historical transition count is

$$N_{\text{past},g}(t, T) = \sum_{j : t - T < t_{g,j} \leq t} H_{g,j}. \tag{6}$$

The past rate is defined using the 60-second count:

$$r_{\text{past},g}(t) = N_{\text{past},g}(t, 60) \quad \text{events/min}. \tag{7}$$

Since the observation window is one minute, the numerical count equals the rate in events per minute.

E. Future Transition-Count Regression

For horizon $T \in \{30, 60\}$ seconds, the target is

$$Y_{g,T}(t) = N_{\text{future},g}(t, T) = \sum_{j : t < t_{g,j} \leq t + T} H_{g,j}. \tag{8}$$

Only observations with a complete future window are retained. The regression model estimates a non-negative conditional count, $\hat{Y}_{g,T}(t)$. A predicted count can be converted to events per minute using

$$\hat{r}_T = \frac{60}{T} \hat{Y}_T. \tag{9}$$

III. Dataset Description

A. Source and Simulation Runs

The data are obtained from the `dataset_labeled_w3.csv` files in the Hugging Face repository `FUMD-AI/krakow-handover-dataset` [3]. Four simulation runs are combined. Each source file contains 37 columns; 28 columns are retained in the working dataframe.

Run-specific vehicle identifiers are not globally unique. A composite group identifier is therefore created by concatenating the run name and vehicle identifier. This prevents identical numeric identifiers from different simulation runs from being treated as the same vehicle.

TABLE II
RUN-LEVEL DATASET STATISTICS

Run	Observations	Vehicles	Source columns
900_1	259,918	775	37
1000_1	307,879	887	37
1200_1	373,688	1,045	37
1400_1	466,786	1,218	37
Combined	1,408,271	3,925	28 retained

TABLE III
MIGRATION-LABEL DISTRIBUTION

Migration value	Count	Percentage
0	1,379,632	97.9664%
1	9,574	0.6798%
2	19,065	1.3538%
Positive warning ($m > 0$)	28,639	2.0336%

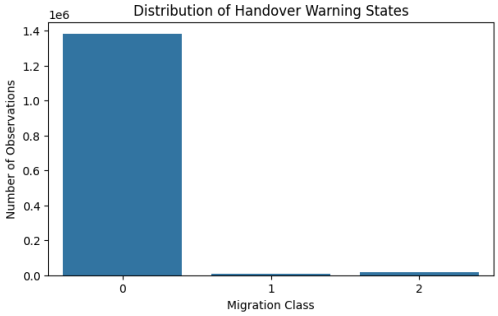


Fig. 1. Distribution of migration labels.

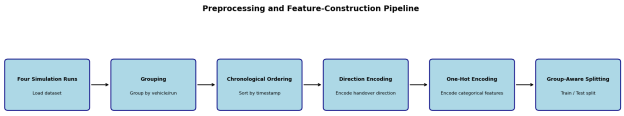


Fig. 2. Preprocessing pipeline: four simulation runs, grouping, chronological ordering, direction encoding, one-hot encoding, and group-aware splitting.

B. Feature Groups

The retained fields include timestamp, vehicle identifier, position coordinates, angle, speed, longitudinal position, signal field, downlink CQI, distance, measured downlink and uplink SINR, received downlink SINR, downlink RLC delay, RLC PDU delay, RLC throughput, current serving cell, seven historical serving-cell fields, migration, destination, run identifier, and group identifier.

The model feature representation uses 15 numerical variables: x, y, *pos*, *speed*, *signals*, angle sine and cosine, average downlink CQI, distance, measured downlink and uplink SINR, received downlink SINR, downlink RLC delay, RLC PDU delay, and downlink RLC throughput. Four serving-cell fields (current serving cell and the first three historical serving-cell fields) are one-hot encoded into 42 indicator columns. The resulting feature matrix has 57 dimensions.

The combined dataframe contains 1,408,271 rows and 28 retained columns. The notebook reports no missing values in these retained columns and no NaN or infinite values in the constructed 57-dimensional feature matrix.

C. Migration-Label Distribution

The migration field is highly imbalanced. Its distribution is given in Table III. The positive warning class is defined as all observations with migration value greater than zero.

Accuracy alone is insufficient for this task. An all-negative classifier can obtain high accuracy while failing to detect any positive observations. Precision-recall metrics, recall, F-scores, and calibration measures are therefore considered.

IV. PREPROCESSING AND FEATURE ENGINEERING

A. Combining Runs and Chronological Ordering

The selected columns from each run are loaded, and the run and composite vehicle-group identifiers are added. The frames are concatenated and sorted by group identifier and timestamp. Within-vehicle timestamp differences are checked to ensure that observations are chronologically ordered.

B. Circular Direction Encoding

Direction is an angular variable. Directly using an angle creates a discontinuity at the wrap boundary: directions close to 0° and 360° are physically close but numerically far apart. The angle θ is therefore represented by

$$\theta_{\sin} = \sin\left(\frac{\pi\theta}{180}\right), \tag{10}$$

$$\theta_{\cos} = \cos\left(\frac{\pi\theta}{180}\right). \tag{11}$$

This representation maps direction onto the unit circle.

C. Encoding and Scaling

The four serving-cell fields are converted to categorical strings and one-hot encoded. The 15 numerical features and 42 indicator columns are concatenated to form the 57-dimensional matrix. For the principal classification pipeline, median imputation and standardization are fitted on the training partition only, then applied to validation and test partitions. The one-hot columns are concatenated without standard scaling.

The temporal pipeline adds three numerical history variables: past 30-second count, past 60-second count, and cumulative transition count. It therefore contains 18 numerical features and 42 one-hot indicators, giving 60 dimensions. Its imputer and scaler are also fitted on training data only.

TABLE IV
DESCRIPTIVE STATISTICS OF DERIVED TEMPORAL QUANTITIES

Quantity	N	Mean	Median	Maximum
Cumulative transitions	1,408,271	4.6802	3	64
Past count, 30 s	1,408,271	0.5647	0	20
Past count, 60 s	1,408,271	1.0678	0	30
Future count, 30 s	1,291,555	0.5830	0	20
Future count, 60 s	1,177,016	1.1313	0	30

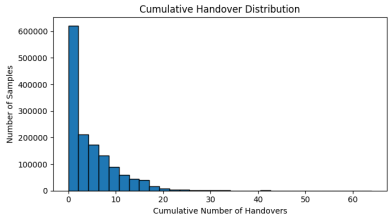


Fig. 3. Distribution of cumulative serving-cell transitions.

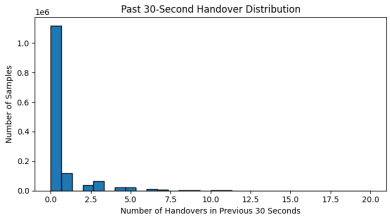


Fig. 4. Distribution of transitions in the preceding 30 seconds.

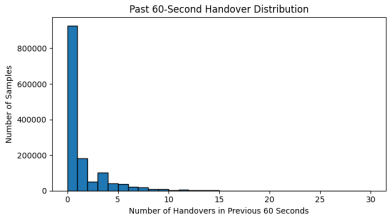


Fig. 5. Distribution of transitions in the preceding 60 seconds.

V. Temporal Target Construction

A. Efficient Interval Counting

For a vehicle, let the sorted event-time vector be $\boldsymbol{e} = (e_1, \ldots, e_M)$. The number of events in an interval is computed using two binary searches. For the past interval $(t-T, t]$,

$$N_{\text{past}}(t,T) = \text{searchsorted}(\boldsymbol{e}, t, \texttt{right}) - \text{searchsorted}(\boldsymbol{e}, t-T, \texttt{right}). \tag{12}$$

For the future interval $(t, t+T]$,

$$N_{\text{future}}(t,T) = \text{searchsorted}(\boldsymbol{e}, t+T, \texttt{right}) - \text{searchsorted}(\boldsymbol{e}, t, \texttt{right}). \tag{13}$$

The right insertion boundary implements the open-left, closed-right interval convention. Each count query is logarithmic in the number of event times.

B. Complete Future Windows

A sample is valid for horizon T only if $t + T \le t_{\max,g}$, where $t_{\max,g}$ is the final timestamp for that vehicle. Otherwise, the future target is marked missing and excluded from training. The 60-second complete-window subset is used for the temporal split, so both 30- and 60-second targets are available for every retained observation.

C. Difference Between Warning and Transition Labels

The notebook reports 28,639 positive binary migration-warning samples, whereas the serving-cell transition detector identifies 27,974 transition events. These counts differ because the two targets have different definitions: migration is a supplied warning label, while a transition is derived from adjacent serving-cell values. They are not treated as interchangeable.

The 30-second future target has 116,716 invalid tail observations, while the 60-second target has 231,255. The longer horizon excludes more observations because a complete future interval is required.

VI. Machine Learning Models

A. Gaussian Naive Bayes

Naive Bayes applies Bayes' rule with conditional independence between features given the class:

$$P(y = c \mid \boldsymbol{x}) \propto P(y = c) \prod_{j=1}^{d} P(x_j \mid y = c). \tag{14}$$

For Gaussian Naive Bayes, each feature likelihood is modelled as $x_j \mid y = c \sim \mathcal{N}(\mu_{cj}, \sigma_{cj}^2)$. The model is computationally efficient, but its conditional-independence assumption can be restrictive when radio and mobility features are correlated.

B. Linear and Quadratic Discriminant Analysis

LDA assumes a Gaussian class-conditional distribution with shared covariance:

$$\boldsymbol{x} \mid y = c \sim \mathcal{N}(\boldsymbol{\mu}_c, \Sigma). \tag{15}$$

Its discriminant function is

$$\delta_c(\boldsymbol{x}) = \boldsymbol{x}^T \Sigma^{-1} \boldsymbol{\mu}_c - \frac{1}{2} \boldsymbol{\mu}_c^T \Sigma^{-1} \boldsymbol{\mu}_c + \log \pi_c. \tag{16}$$

The shared covariance produces linear decision boundaries. The implementation uses the least-squares solver with automatic shrinkage.

QDA instead estimates a separate covariance matrix for each class:

$$\boldsymbol{x} \mid y = c \sim \mathcal{N}(\boldsymbol{\mu}_c, \Sigma_c). \tag{17}$$

Its discriminant is

$$\delta_c(\boldsymbol{x}) = -\frac{1}{2} \log |\Sigma_c| - \frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_c) + \log \pi_c. \tag{18}$$

QDA permits quadratic decision boundaries. A regularization parameter of 0.2 is used to stabilize covariance estimation.

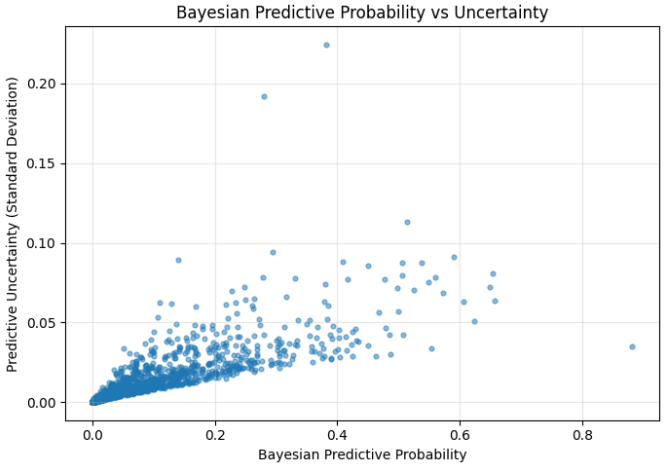


Fig. 6. Bayesian logistic predictive probability versus sampled predictive uncertainty.

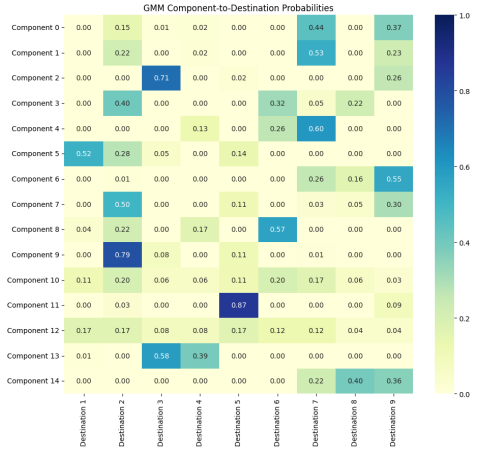


Fig. 7. Estimated destination probability for each GMM component.

C. Logistic Regression

For binary classification, logistic regression estimates

$$P(y = 1 \mid \boldsymbol{x}) = \sigma(\boldsymbol{w}^T \boldsymbol{x} + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{19}$$

The negative log-likelihood is the binary cross-entropy:

$$\mathcal{L} = -\sum_i \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]. \tag{20}$$

The implementation uses the LBFGS solver, a maximum of 300 iterations, random state 42, and balanced class weights. For destination prediction, the model estimates multiclass probabilities and selects the class with maximum probability.

D. Bayesian Logistic Regression

A zero-mean Gaussian prior is placed on the weight vector, excluding the intercept:

$$p(\boldsymbol{w}) \propto \exp\left(-\frac{\alpha}{2} \sum_{j=1}^{d} w_j^2 \right), \qquad \alpha = 1. \tag{21}$$

The maximum-a-posteriori (MAP) estimate minimizes

$$\mathcal{J}(\boldsymbol{w}) = \sum_i \left[\log(1 + e^{\boldsymbol{w}^T \tilde{\boldsymbol{x}}_i}) - y_i \boldsymbol{w}^T \tilde{\boldsymbol{x}}_i \right] + \frac{\alpha}{2} \sum_{j=1}^{d} w_j^2, \tag{22}$$

where $\tilde{\boldsymbol{x}}$ includes the intercept. The implementation optimizes this objective using L-BFGS-B. A Laplace approximation represents the posterior near the MAP estimate as

$$p(\boldsymbol{w} \mid D) \approx \mathcal{N}(\hat{\boldsymbol{w}}_{\text{MAP}}, H^{-1}), \tag{23}$$

where H is the Hessian of the negative log posterior at the MAP estimate. One hundred weight vectors are sampled from this approximate posterior, and their sigmoid predictions are averaged. The standard deviation of sampled probabilities is used as a predictive-uncertainty measure.

E. Gaussian Mixture Model

A GMM represents the feature density as a weighted sum of Gaussian components:

$$p(\boldsymbol{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_k, \Sigma_k), \quad \sum_{k=1}^{K} \pi_k = 1. \tag{24}$$

The posterior responsibility of component k for observation i is

$$r_{ik} = \frac{\pi_k \mathcal{N}(\boldsymbol{x}_i \mid \boldsymbol{\mu}_k, \Sigma_k)}{\sum_{\ell=1}^{K} \pi_\ell \mathcal{N}(\boldsymbol{x}_i \mid \boldsymbol{\mu}_\ell, \Sigma_\ell)}. \tag{25}$$

Expectation-Maximization alternates between responsibility calculation and parameter updates. For diagonal covariance, only feature-wise variances are retained.

The binary GMM uses eight components, diagonal covariance, covariance regularization 10^{-4}, at most 100 iterations, and a random 50,000-row training subset. Component-level warning probabilities use additive smoothing:

$$\hat{q}_k = \frac{\sum_i r_{ik} y_i + 1}{\sum_i r_{ik} + 2}, \qquad \hat{P}(y = 1 \mid \boldsymbol{x}) = \sum_k r_k(\boldsymbol{x}) \hat{q}_k. \tag{26}$$

For destination prediction, the GMM is fitted to warning-positive observations after zero-variance filtering and standardization. Candidate component counts are $\{2, 3, 4, 5, 6, 8, 10, 12, 15\}$, and the selected count minimizes validation log loss. The final model uses 15 components, diagonal covariance, covariance regularization 0.01, five initializations, and at most 300 iterations. Component-to-destination probabilities are estimated using one-hot destination labels and additive smoothing:

$$P(d = c \mid z = k) = \frac{\sum_i r_{ik} \mathbb{1}(d_i = c) + \alpha}{\sum_i r_{ik} + \alpha |\mathcal{C}|}, \quad \alpha = 1. \tag{27}$$

The final destination probability is the responsibility-weighted mixture over components.

TABLE V
Primary classification data split

Partition	Rows	Vehicle groups
Training (before cap)	920,573	2,551
Validation	204,782	589
Test	282,916	785

F. Poisson Regression

For a count response, Poisson regression assumes

$$Y_i \sim \text{Poisson}(\lambda_i), \qquad P(Y_i = y_i) = \frac{e^{-\lambda_i}\lambda_i^{y_i}}{y_i!}, \tag{28}$$

with a log-linear mean

$$\log \lambda_i = \boldsymbol{w}^T \boldsymbol{x}_i + b. \tag{29}$$

The log link ensures positive predicted means. The implementation uses PoissonRegressor with $\alpha = 1$, a maximum of 300 iterations, and tolerance 10^{-5}.

G. Histogram Gradient Boosting

Gradient boosting constructs an additive predictor

$$F_M(\boldsymbol{x}) = F_0(\boldsymbol{x}) + \sum_{m=1}^{M} \eta h_m(\boldsymbol{x}), \tag{30}$$

where each tree is fitted to improve the current loss. With Poisson loss, the model learns nonlinear relationships between features and the conditional event-count mean. The implementation uses learning rate 0.08, 150 boosting iterations, at most 31 leaf nodes per tree, L2 regularization 1.0, and random state 42.

VII. Experimental Methodology

A. Vehicle-Group-Aware Classification Split

The primary classification split uses two GroupShuffleSplit operations with random state 42. First, 20% of vehicle groups are held out for testing. The remaining groups are divided into training and validation partitions, corresponding to 65%, 15%, and 20% of the overall data. No vehicle group is shared between partitions.

The binary training set is reduced to 300,000 observations using stratified random sampling. It contains 293,899 negative and 6,101 positive samples. Validation and test sets are not downsampled. The binary GMM uses 50,000 training samples, while Bayesian logistic regression uses 80,000 samples and 100 approximate posterior samples.

B. Destination and Temporal Splits

Destination classification uses warning-positive subsets containing 6,101 training, 4,108 validation, and 5,809 test observations. For count regression, only rows with complete 60-second future targets are retained. A separate group-aware split produces 763,571 training, 175,661 validation, and 237,784 test rows. Each temporal regressor is fitted using a random 300,000-row subset of the training partition.

TABLE VI
Validation threshold selection for binary prediction

Model	Threshold	Validation F2
Gaussian Naive Bayes	0.87	0.2305
LDA	0.05	0.3677
QDA	0.03	0.4346
Logistic Regression	0.65	0.4139
GMM	0.04	0.1783
Bayesian Logistic Regression	0.06	0.0000

C. Threshold Selection

For each binary model, 99 thresholds between 0.01 and 0.99 are evaluated on the validation set. The threshold maximizing validation F2-score is selected and applied unchanged to the test probabilities. F2 gives greater weight to recall than precision and is useful when missed warning events are considered more costly than false alarms.

D. Evaluation Metrics

Binary classification is evaluated using accuracy, precision, recall, F1-score, F2-score, ROC-AUC, PR-AUC, and Brier score. Destination classification is evaluated using accuracy, balanced accuracy, macro precision, macro recall, macro-F1, and log loss. Future-count regression is evaluated using mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and Poisson deviance.

VIII. Results and Analysis

A. Binary Migration-Warning Prediction

The validation-selected thresholds are shown in Table VI. Bayesian logistic regression has a validation F2-score of zero at its selected threshold, despite achieving the highest test PR-AUC. This discrepancy indicates potential instability or a validation/test distribution difference and requires further investigation.

Bayesian logistic regression achieves the highest PR-AUC (0.2194), while QDA obtains the highest F1-score (0.2685) and F2-score (0.4348). Logistic regression gives the highest recall (0.8370) and ROC-AUC (0.9399). The model selected for the displayed confusion matrix is Bayesian logistic regression because it has the highest PR-AUC.

The number of false positives is substantially larger than the number of true positives, explaining the low precision despite considerable recall. The all-negative baseline obtains approximately 97.95% accuracy but has zero recall and zero F1-score. This confirms that accuracy is misleading for the 2.03% positive-class prevalence. The mean predictive uncertainty reported for Bayesian logistic regression is 0.002951; a dedicated calibration study is still required to establish whether this uncertainty is reliable.

B. Destination Serving-Cell Prediction

Destination prediction is conditioned on a positive migration warning. The test results are given in Table VIII.

TABLE VII
BINARY HANDOVER-WARNING PREDICTION: TEST-SET PERFORMANCE

Model	Accuracy	Precision	Recall	F1	F2	ROC-AUC	PR-AUC	Brier
Bayesian Logistic	0.9146	0.1541	0.7043	0.2529	0.4109	0.9370	0.2194	0.0176
LDA	0.9050	0.1322	0.6524	0.2199	0.3652	0.8943	0.2173	0.0182
QDA	0.9171	0.1639	0.7411	0.2685	0.4348	0.9367	0.2158	0.0188
Logistic Regression	0.8911	0.1400	0.8370	0.2398	0.4193	0.9399	0.2013	0.1097
Gaussian Naive Bayes	0.8589	0.0718	0.4923	0.1253	0.2267	0.8076	0.0748	0.1601
GMM	0.8410	0.0545	0.4125	0.0963	0.1783	0.7554	0.0720	0.0193

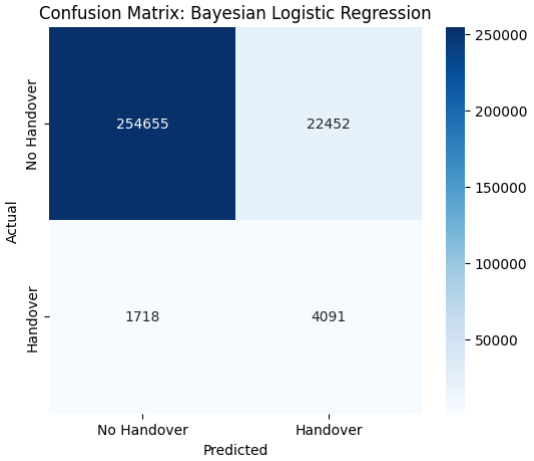


Fig. 8. Test confusion matrix for Bayesian logistic regression. Recorded counts: TN=254,653, FP=22,454, FN=1,718, TP=4,091.

TABLE VIII
DESTINATION CLASSIFICATION: TEST-SET PERFORMANCE

Model	Acc.	Bal. Acc.	Macro P.	Macro R.	Macro-F1	Log loss
Logistic Regression	0.8965	0.9196	0.8887	0.9196	0.9024	0.2721
QDA	0.8544	0.9003	0.8529	0.9003	0.8668	0.4289
LDA	0.8411	0.8469	0.8348	0.8469	0.8395	0.5218
Gaussian Naive Bayes	0.5269	0.6763	0.5824	0.6763	0.5453	13.4219
GMM	0.5627	0.5406	0.4990	0.5406	0.4919	1.0045

Logistic regression is the strongest model across the displayed metrics, obtaining macro-F1 of 0.9024 and log loss of 0.2721.

The GMM destination model selects 15 components by validation log loss and converges in 14 iterations. It reports AIC of −100087.2152 and BIC of −88508.4727. Its test top-1 accuracy is 0.5627 and macro-F1 is 0.4919, whereas top-3 accuracy is 0.9726. The difference indicates that the probability ranking often includes the correct destination among the three candidates, but top-3 accuracy must not be interpreted as top-1 correctness.

C. Future Handover-Count Prediction

The temporal regression models use the original 57 features and three historical variables: past 30-second count, past 60-second count, and cumulative transition count. The two horizons are trained separately. Table IX presents the recorded results.

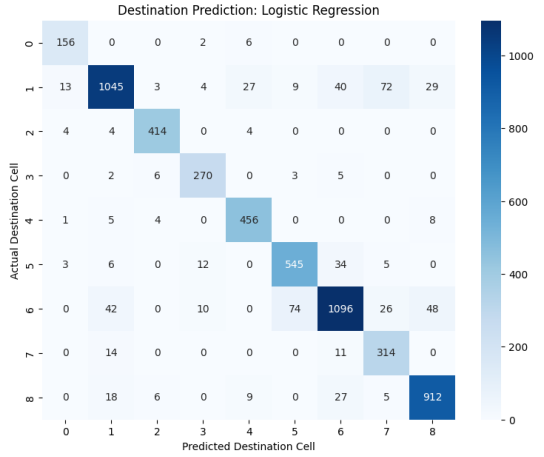


Fig. 9. Destination confusion matrix for logistic regression.

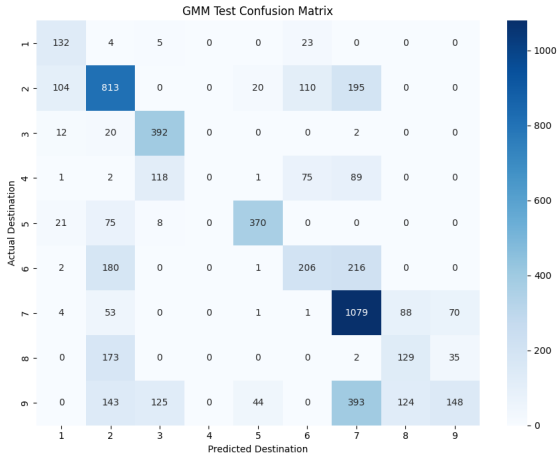


Fig. 10. GMM destination test confusion matrix.

Histogram gradient boosting outperforms Poisson regression for both horizons on all reported test metrics. Its test MAE increases from 0.4261 at 30 seconds to 0.8012 at 60 seconds, consistent with the larger number of transitions that can occur over a longer interval. Its R^2 remains above 0.52 for both horizons. The weaker Poisson regression results suggest that a linear log-mean relationship does not fully represent the feature-target relationship.

For a 60-second horizon, predicted count and predicted events per minute have the same numerical value. For a

TABLE IX
FUTURE HANDOVER-COUNT REGRESSION RESULTS

Horizon	Model	Val. MAE	Test MAE	RMSE	R^2	Dev.
30 s	Poisson	0.7532	0.7493	1.3152	0.1749	1.4792
30 s	Hist. Grad. Boost.	0.4198	0.4261	0.9659	0.5549	0.6402
60 s	Poisson	1.2509	1.2433	1.9267	0.1935	2.1476
60 s	Hist. Grad. Boost.	0.7947	0.8012	1.4757	0.5269	1.1086

30-second horizon, the predicted count is multiplied by two to express an events-per-minute rate. This conversion is not a separately trained model.

IX. Discussion

The best model depends on the task and evaluation criterion. Bayesian logistic regression leads in binary PR-AUC, QDA leads in binary F2 and F1, logistic regression leads in destination macro-F1, and histogram gradient boosting leads in future-count regression. No single model is optimal for all objectives.

The binary warning target is highly imbalanced. Models that increase recall can generate many false positives, so operational selection requires an explicit cost function that accounts for missed handovers, unnecessary handover preparation, signalling overhead, and service interruption. PR-AUC is useful under class imbalance, but thresholded precision and recall and probability calibration are also necessary for selecting an operational policy.

The GMM provides a latent-component representation, but its current results do not outperform direct discriminative classification. The large difference between top-3 and top-1 destination accuracy suggests that its ranking may still provide useful candidate information, although probability calibration and decision policy require further evaluation.

A. Limitations

Several limitations should be considered before interpreting the results as a deployable 5G mobility solution:

1) **Timestamp units:** the implementation sets `TIME_TO_SECONDS = 1.0`, assuming that the timestamp is measured in seconds. This assumption must be verified from dataset metadata before the 30- and 60-second targets can be treated as confirmed real-time horizons.
2) **Different event definitions:** migration-positive rows and serving-cell transitions are derived differently and have different counts.
3) **Destination conditioning:** destination metrics apply only to warning-positive observations.
4) **Feature availability:** current and historical serving-cell fields must be available at the intended prediction instant and must not encode post-decision information.
5) **Sampling:** different models use different training subsets, so results depend on the implemented sampling choices.
6) **Validation-threshold anomaly:** Bayesian logistic regression has validation F2 equal to zero at the selected threshold despite leading test PR-AUC. The validation probability distribution and threshold behaviour require investigation.
7) **No sequence architecture:** the temporal regressors use tabular features and engineered history counts rather than recurrent or transformer-based sequence models.
8) **Calibration:** Brier score is reported, but reliability diagrams and formal calibration correction are not included.
9) **External validation:** all experiments use four simulation runs from one dataset; generalization to other cities, traffic conditions, radio configurations, or real network traces is not established.

X. Conclusion and Future Work

This paper presents a multi-objective machine learning pipeline for 5G handover analysis using four Krakow simulation runs. It separates the migration-based binary warning label from the serving-cell transition process and uses the latter to derive cumulative counts, historical windows, and future count targets. The results show that model choice should be task-specific: Bayesian logistic regression achieves the highest binary PR-AUC, QDA the highest binary F2-score, logistic regression the highest destination macro-F1, and histogram gradient boosting the strongest future-count regression performance.

The mean past 60-second transition count is 1.0678 events per minute, while the mean future 60-second target is 1.1313. Histogram gradient boosting substantially reduces future-count MAE relative to Poisson regression, indicating that nonlinear relationships among mobility, radio, and network measurements are relevant.

Future work should first verify timestamp units and audit feature availability at prediction time. Further directions include probability calibration, cost-sensitive threshold selection, explicit modelling of handover failures and ping-pong events, temporal sequence models, spatial trajectory features, uncertainty-aware destination ranking, external validation on independent network traces, and evaluation under realistic inference-latency constraints.

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